Drug Overdose and Abuse in Communities

Andrea Caceres
Isra Marcu
Amber Teetsel

University of Colorado Boulder
April 13, 2026

Drug Overdose and Abuse in Communities

Andrea Caceres
anca9584@colorado.edu
University of Colorado Boulder
Boulder, Colorado, USA

Isra Marcu
isma5785@colorado.edu
University of Colorado Boulder
Boulder, Colorado, USA

Amber Teetsel
amber.teetsel@colorado.edu
University of Colorado Boulder
Boulder, Colorado, USA

Abstract

Drug abuse has cascading effects across multiple industries, affecting a large variety of people. Although drug abuse has been researched in the past there are many pressing questions that need to be explored further. We are interested in understanding who drug use affects and when it affects them. Additionally, it is important to look at drug use intersectionality, thinking about mental health and various socioeconomic factors that impact this use. Datasets used for this analysis include NCHS Drug Poisoning Death Mortality Rate, Accidental Drug Related Deaths, and Treatment Episode Data Set: Admissions.

1 Introduction

The topic of our research project is predicting which communities are at a higher risk of drug abuse and or drug overdose. By using socioeconomic, demographic, drug type, and past drug overdose data, we can help identify communities with high rates of substance abuse. Once these communities are identified, we will analyze the underlying factors that contribute to this elevated risk. Some of these factors could be housing insecurity, food insecurity, age, genetics, location, exposure to prescriptions drugs, or access to treatment.

This research topic is important because substance abuse is alarmingly prevalent in the United States. Drug overdose is the leading cause of death for Americans under 50 years old (Substance Abuse and Mental Health Services Administration, 2025). In 2024, only 1 in 5 people who suffer from drug abuse received treatment (SAMHSA, 2025). Among people in 2024, 71.5% of adolescents aged 12 to 17 years used nicotine products in the past month (SAMHSA, 2025).

If we could help predict which communities are at risk and why, it could help distinguish where substance abuse treatment resources need to go and identify which factors that lead to substance abuse. This research benefits treatment providers, schools and community organizations, and public health agencies.

1.1 Significance and Stakeholders

1.1.1 Stakeholders. Drug abuse has cascading effects across multiple industries. Real-world implications of substance abuse are intergenerational substance abuse, increased taxes and spending to keep up with this crisis, and entrenched poverty. This research could help healthcare and public health industries take a preventative approach to substance abuse rather than a reactive one. By using a data-supported stratification rather than minuscule intervention, the healthcare system can be more effective in their approach to fighting drug abuse. Predictive modeling is beneficial because it provides the most optimized approach to using limited and minimal resources. This ensures that the taxpayer dollar is utilized in an efficient approach towards prevention of drug addiction. This research could help lower the volume of non-violent offenders in jails and prisons. Schools and community organizations will improve if preventative measures can be taken because drop-out rates will decrease. Also, teachers and professors will have identifiers for students and or participants for risk of drug abuse. Not only can patterns be identified early, but schools can become a safe space for students to be vulnerable psychologically. This can allow them to focus on their academics rather than stressors from their environment outside of school. Marginalized populations could benefit from this research because resources and treatment can be focused on these groups if they are identified, setting up these communities to succeed. If these issues can be addressed throughout our research, the loop of generational poverty can be broken. The lack of education amongst marginalized groups can also be fixed as substance abuse decreases and more opportunities can arise. Breaking these trends can increase social equity. Plus, this research can provide the means necessary to inflict systemic changes in society. Legislation can be made to implement the results these types of studies produce in aiding those struggling. Bringing down the substance abuse crisis in the country can improve quality of life for individuals in major cities as well. Streets will be cleaner, the poverty rate will lower, unemployment rate will decrease, and these communities can produce contributing citizens. Identifying these trends can help prevent drug abuse and increase the professional integrity of those more likely to be at risk.

1.1.2 Existing Solutions & Gaps. There are many imperfect solutions to combat drug overdose and help mitigate drug abuse. In specific there are opioid overdose reversal medications (OORM) that are life-saving and reverse the effects of opioid overdose (Substance Abuse and Mental Health Services Administration, 2023). These medications are widely available for the general public and do not require training, significantly reducing overdose fatality rates (SAMHSA, 2023). Another way to reduce the risk of overdose is using test strips to check for the presence of fentanyl (SAMHSA, 2023). In general effective treatment of substance use disorder can reduce the risk of overdose and help combat drug use, whether that be through out-patient therapy or in-patient stay at a facility. There are many challenges regarding drug use; one is the inappropriate and over-prescribing of opioids, leading to misuse (Cerdá, Krawczyk, Keyes, 2023). In addition, the criminalization of drug use has harmed those that are users and/or in possession of drugs resulting in incarceration rather than treatment (Cerdá, Krawczyk, Keyes, 2023). Many challenges occur because of a lack of resources, such as poor access to healthcare to address pain treatment (SAMHSA, 2023). To truly prevent overdose there needs to be steps taken before overdose can even occur to help mitigate the risk. This can be done by assessing personal risk and taking the steps that are specific to

those factors because treatment needs to be individualized to truly be effective.

1.2 Research Questions

(1) What are the most common circumstances of death among substance users?
(2) How do drug profiles (mix of drugs in system) differ between overdose victims?
(3) How has the age of first use changed over time for different drugs?
(4) Does time of year affect overdose deaths? If so, how does the seasonality of overdose deaths differ between drugs?
(5) Which socioeconomic factors have the strongest correlation with overdose?
(6) How do mental health and substance abuse intersect? Does this differ by gender or race?
(7) How does forced vs. involuntary substance abuse treatment impact treatment completion rates?
(8) Can intake demographics allow us to predict whether substance abuse treatment will be completed or not?
(9) What demographic features are the best predictors of poly-substance mortality?
(10) Can we forecast the next XX months of overdose death for specific drugs?

2 Data Description

2.1 NCHS Drug Mortality

2.1.1 Overview. This dataset comes from the Centers for Disease Control (CDC) and National Center for Health Statistics (NCHS). It is accessed via a SODA3 API. The dataset provides solid historical information (1999-2016) on drug deaths in the United States broken down by State, Age Group, Race (White, Hispanic, Black), and Sex (Male, Female). It enables us to see trends in drug mortality during the early 21st century and can be useful in building an eventual predictive model.

2.1.2 Data Cleaning and Pre-processing.

Handling Issues/Noise in the Data. The dataset was missing values related to age-adjusted death rate for all rows where age group was specified (i.e. not "All Ages"). Age-adjusted rates are only applicable for the "All Ages" category, so we substituted the crude rate in a new combined column called `death_rate` and dropped the old age-adjusted rate column.

We dropped the `state_crude_rate_in_range_column` because the values were overlapping/inconsistent and it can be easily recreated to our specifications if needed.

We identified many high outliers for death rate but decided to leave them in the data. Upon inspection, the numbers are consistent with the raw deaths/population numbers reported and tend to occur in states where we know the opioid and fentanyl crises had devastating effects (West Virginia, Ohio, etc.). Thus we feel the outliers are useful data points for understanding the true substance abuse landscape in the United States.

Understanding the Data. All data was initially imported as object type. We converted year, population, deaths, and all death rate columns to numeric. Categorical columns include state, race, sex, and age group. We applied z-score standardization and a log transformation to death rate for use in future modeling.

Statistical Analysis. Summary statistics were computed in Python for all numeric columns. We also found the death rate is right-skewed, reflecting a few state "hotspots" with high death rates. Details can be found on project website.

Data Quality. After cleaning, this data is extremely usable, consistent, and complete. There are no missing values, and every year contains consistent types of information at cumulative and de-aggregated levels. Additionally, reporting the age-adjusted death rate per 100,000 population ensures consistency between states with different populations and demographics. The most important thing to keep in mind when using this dataset is the need to filter before specific analysis because it contains both cumulative and granular rows (e.g. data for each year contains information for all 50 states, plus a row for the United States as a whole. Similar structure for Sex, Race, and Age Group). To this end, we've created some subsets - one for the completed aggregated data (US-level, all ages, both sexes, all races) and for state-level data (all ages, both sexes, all races).

Advanced Data Understanding. QQ Plots demonstrated that death rate is approximately normal, despite some outliers on the high end.

2.1.3 Biases and Limitations. The dataset may under-report drug death. It relies on death certificates which may be subject to reporting bias (under-reported due to stigma). It also ends in 2016 meaning it doesn't provide insight to the past decade of drug use and overdose in the United States.

2.2 Accidental Drug Related Deaths in Connecticut from 2012-2024

Overview To find the Accidental Drug Related Deaths 2012-2024 for the state of Connecticut, I utilized data.gov, a government portal, to find this data. Data.gov is a highly authoritative source for U.S. local, federal, and local data. I pulled the API for this dataset from data.ct.gov. This dataset was last updated September 12, 2025 from the Office of the Chief Medical Examiner. This dataset is relevant to our questions because this dataset contains context behind the overdose-related deaths in Connecticut. It also contains the drugs involved in each death. Various columns in this dataset are different drugs, the rows contain 'Y' (involved in cause of death) and 'N' for no. Using these columns can allow us to determine which drugs or mixture of drugs is leading overdose deaths in Connecticut. Each row is a death that has its date captured, therefore we can delve into if time of year has an effect on overdose amounts and if so, what is the pattern? This dataset can help us answer many of our questions proposed for our project.

2.2.1 Data Cleaning and Pre-processing.

Handling Issues/Noise in the Data. In the dataset, there were columns such as 'injurycity', 'deathcity', and 'residencecity' that I removed because there were >500 cities listed and around $\tilde{2}00$ counties. I determined that if clustering was conducted in the future,

'Residence County'] would be easier to handle for this method. Columns that contained coordinates such as 'residencecitygeo', 'deathcitygeo', and 'injurycitygeo' were removed since we will not be conducting any spatial analysis. The column 'datetype' was removed because it did not provide any potential analytical insight. I removed all columns that were missing 50% or more of the data because these columns lacked the sufficient amount of information to be useful in analysis. For smaller amounts of missing values, there were many methods used to clean this dataset. For the 'Age' column, I replaced missing values with the median age which is 44 years old. The reason why I used the median and not the mean is because the median is the least affected by outliers and therefore the data will not be as skewed. For my categorical variables, such as, 'Race', and 'Location', I filled the missing values with 'Unknown' to make analysis easier. The 'Sex' column had two types of data besides 'Male' and 'Female': 'Unknown' and 'X'. For these entries, they were converted to NaN to improve accuracy since 'Sex' is a binary categorical variable. Then, any duplicate records were located and removed from the data, to ensure that each case of an overdose only shows up once in the data. There were many categorical variables with typos or inconsistent formatting. For example, in 'Race' there would be an unique response 'white' and a separate unique response 'White'. This standardization was conducted for multiple variables. Another standardization done was for capitalization and spacing. The data columns from the raw dataset were lowercase and did not have spacing in between words. After fixing this, the column 'fentanylanalogue' changed to 'Fentanyl Analogue'. I also grouped different responses into one category. For example, for 'Location' of where the accidental drug related death occurred, there were responses like: 'Decadant's Home' and 'Residence'. These were changed to just 'Home' to improve the consistency. Outliers in the data for 'Age' were removed by using thresholds for ages less than 10 and older than 100.

Understanding the Data. The dataset had different types of variables: numerical variables (Age, Drug Count), categorical variables (Sex, Race, Location, Manner of Death, County) and binary variables (Fentanyl, Cocaine, Heroin, etc.), and time variables (Date). For additional trend analysis over time, I created two variables, 'Year' and 'Month' to determine the seasonality of accidental drug related deaths over time in Connecticut. Another column, 'Drug Count' was created to determine how many drugs were used in poly-drug overdoses and which count is the most common. For another drug indicator, I changed the responses for each column drug from 'Y' and 'N' to '1' and '0', making the column a binary integer feature. Transformations were also made to ensure a more efficient analytical process. For example, using z-score normalization, 'Age' was transformed into a standardized scale with mean 0 and variance of 1 and turned into a new column, 'Age Standardized'. 'Drug Count Log' was created to convert 'Drug Count' with a logarithmic transformation to reduce skewness in the drug count distribution.

Statistical Analysis. Figure 1 contains a table with the summary statistics information. For correlation analysis, a heatmap visualization was made using the Seaborn package. This was able to help identify which drugs were common in drug combinations for overdose cases.

	Mean	Median	Variance	Standard Deviation	Skewness	Min	Max
Age	44.35439327	44	161.48217	12.70756359	0.04121917217	13	87
Heroin	0.2804906272	0	0.2018312	0.4492562799	0.9773630551	0	1
Cocaine	0.3983645761	0	0.2396887	0.4895801577	0.4152582802	0	1
Fentanyl	0.6797037723	1	0.2177233	0.4666083475	-0.7703741399	0	1
Fentanyl Analogue	0.0787626321	0	0.0725646	0.2693783171	3.127960507	0	1
Oxycodone	0.0825426213	0	0.0757351	0.2752002532	3.034312587	0	1
Oxymorphone	0.0134999614	0	0.0133187	0.1154068452	8.432344012	0	1
Ethanol	0.2570392656	0	0.1909848	0.4370180941	1.112073979	0	1
Hydrocodone	0.0133456761	0	0.0131685	0.114754455	8.482972082	0	1
Benzodiazepine	0.2245622155	0	0.1741474	0.4173097903	1.320267549	0	1
Methadone	0.0918768803	0	0.0834419	0.2888632135	2.826154925	0	1
Methamphetamine	0.0097971148	0	0.0097018	0.09849812097	9.955083875	0	1
Amphetamine	0.0381856051	0	0.0367302	0.191651502	4.820057784	0	1
Tramadol	0.0239142173	0	0.0233441	0.1527878543	6.232946318	0	1
Hydromorphone	0.0061714109	0	0.0061337	0.07831856615	12.61271281	0	1
Morphine	0.0045514155	0	0.0045310	0.06731307244	14.72299566	0	1
Xylazine	0.1094654015	0	0.0974902	0.3122342839	2.501934487	0	1
Gabapentin	0.0427370207	0	0.0409137	0.2022714117	4.52197901	0	1
Opiate NOS	0.0098742575	0	0.0097775	0.09888129653	9.914946315	0	1
Heroin Morph Codeine	0.1711023683	0	0.1418372	0.3766129174	1.74687786	0	1
Year	2019.155905	2020	10.80665912	3.287348341	-0.3979255971	2012	2024
Month	6.559824115	7	11.6881929	3.418799921	-0.01193216057	1	12
Drug Count	3.289130602	3	2.1321044	1.460172763	0.3356617437	0	10
Age Standardized	2.06E-16	-0.0278894	1.0000771	1.000038574	0.04121917217	-2.467475572	3.356052593
Drug Count Log	1.391557306	1.3862943	0.1410999	0.3756327372	-0.7569328755	0	2.397895273

Figure 1: Summary Statistics Table

Data Quality. Standardizing columns and fixing typos in responses helped aid in consistency for data completeness. To improve data quality and ensure usability, I removed all columns that were missing 50% or more of the data, such as, 'locationifother', 'othersignifican', and 'heroin_dc'.

Advanced Data Understanding. For the advanced data understanding, I wanted to explore the distribution of drug counts versus log drug counts to help evaluate skewness. For normality testing, I plotted a Q-Q plot to see if the data points of ages follow a normal distribution to then determine where parametric statistical methods in the future are applicable.

2.2.2 Biases and Limitations. There are many ethical considerations that apply to this dataset because it contains very sensitive public health information involving deaths caused by overdoses. Another limitation is that there might be reporting biases amongst races and ethnicities. Plus, there are columns with several missing responses which may affect analysis and conclusions.

2.3 Treatment Episode Data Set: Admissions

2.3.1 Overview. The treatment Episode Data Set: The Admission Data Set (TEDS-A) contains demographic and substance use characteristics of people that have been admitted to substance use treatment. This allows the analysis of state-licenses or certified

substance use treatment centers that receive federal funding. This dataset comes from the Substance Abuse and Mental Health Service Administration (SAMHSA) covering data from 2023. There are 1,625,833 respondents, enabling us to see demographics and treatment information for a wide variety of patients in 2023.

2.3.2 Data Cleaning and Pre-processing.

Handling Issues/Noise in the Data. This dataset is very organized, however it contained columns that were redundant and not useful for further computations. One of the most important steps was changing all -9 values to NaN based upon the code-book. This allowed columns with 50% missing data or more to be dropped. Next, all duplicates were dropped to help with accuracy of the data. We then dropped all rows that were missing data for the critical variables with exclusion to first use, as this had too many missing data points that would interact with the dataset as a whole. The column of first use is important, therefore a separate data frame was dropping all NaN values. For human-readability names of columns were changed.

Recoding. Although the data is coded very nicely to begin with, recoding was done to map numeric codes to human-readable labels for age, sex, race, substances, DSM diagnoses, and other categorical features. This created new columns that can be used for visualizations. In addition, the variables of this dataset are on a small standardized scale however data transformation was applied to 'prior tx' and 'arrests 30days' using log1p. This was helpful as both of those variables are extremely skewed.

Understanding the Data and Statistical Analysis. Outlier detection was run on selected columns and although there are many outliers for columns such as 'arrests 30days' and 'self help 30days', these outliers are not due to issues in data collection but true outliers due to differences in the population. When summary statistics were done age and education showed to be both roughly symmetric. Most of the variables are mildly skewed however there are some extremely skewed such as 'arrests 30days' and 'self help 30days', which is consistent with the outlier detection. The correlation matrix showed that primary substance and route to treatment are positively correlated.

Data Quality. This data is very organized and is complete. Before cleaning the data, it was still usable however not very intuitive. After cleaning was complete this data is easy to use and has great human-readability. This data is not available via API, however it does come from a government website. If this data were available in an API it would not be feasible to use as this dataset is very large.

Advanced Data Understanding. The data does not look normal on the Q-Q Plot however it is not continuous so this makes sense and is not a red flag.

2.3.3 Biases.

Main limitations to this dataset are potential biases from systemic prejudice, affecting minority groups. Additionally because this data is so structured there are many responses that could have been marked as NaN if the participant did not fit a specific category given. This data is not longitudinal, not allowing for analysis over time.

3 Modeling Strategy

3.1 Clustering U.S. States

3.1.1 K-Means Algorithm Selection.

A clustering algorithm was chosen to identify states with similar characteristics and evaluate trends in death rate from drug overdose. The K-Means algorithm won out over DBSCAN and Hierarchical Clustering due to its simplicity and superior performance during basic tests of all three model types.

Assumptions. The K-Means algorithm assumes that clusters are spherical and isotropic. In general, the clusters generated violate this assumption due to spread of data, but not to an extreme degree based on their Silhouette Scores. Another assumption is equal variance of clusters, which again is violated by our data. The third assumption is that clusters are similar in size, which does not hold true for our model. In summary, K-Means assumptions were violated; however, our data meets the assumptions for K-Means better than the assumptions for DBSCAN and Hierarchical Clustering.

Hyperparameter Tuning. The most significant hyperparameter for K-Means is k, the number of clusters. Optimal value of k was initially assessed using the Elbow Method and the Davies-Bouldin (DB) Index. The Elbow Method suggested that either $k = 3$ or $k = 4$ was appropriate, and Silhouette Scores indicated better performance for $k = 4$. The DB Index yielded a lower (better) value for $k = 3$, so we looked at the difference in clusters for both k values to make a final determination. Using $k = 4$ clusters added significant qualitative insight; we were able to determine the difference between states driven by prescription opioid supply (Oxycodone and Hydrocodone) versus states driven by Fentanyl supply. Due to the importance of Fentanyl to the current opioid crises, we felt the value of such qualitative insights outweighed the drop-off in mathematical metric quality.

Insights. The clustering model enabled classifying states into four distinct categories: Low Risk, Moderate Risk, High Risk (Prescription-Driven), and Acute Risk (Fentanyl-Driven). Because we had complete data from 2000-2016, we were also able to visualize and track the progression of the opioid crisis over time.

3.2 Connecticut Drug Related Deaths 2012-2024

3.2.1 Apriori Algorithm Selection.

The apriori algorithm model was chosen for drug related deaths in Connecticut from 2012-2024 because the goal is to determine which drugs were most used in a combination resulting in death. This will provide insight into the circumstances surrounding these deaths and determine which drugs are frequently used together. Frequent-pattern growth is the best model for this because in this case our "market basket" will be the death's related toxicology report.

Assumptions. The assumptions for a frequent-pattern growth model is that the data must be categorical or boolean. We have met this assumption by converting the drug count into a boolean factor so our data will show "True" if the drug was present and "False" if not. Another apriori algorithm assumption is the downward closure property. If an item set of drugs is frequent, then those individual drugs must also be frequent, which in this case is true.

Hyperparameter Tuning. For hyperparameter tuning, we set the minimum support threshold to be 10% and the minimum confidence threshold to be 70%, ensuring that the results being shown have a high probability of being true. To determine the performance evaluation of the apriori algorithm, the support, confidence and lift tell us which drugs and drug combinations were most frequent. If the results yield a value greater than 1 for the lift, then we know whether a certain combination occurs more frequently than it would by random chance.

Solutions. Our apriori analysis shows that fentanyl had a support value of approximately 0.679, meaning that nearly 68% of drug related deaths in Connecticut from years 2012-2024 were caused by fentanyl. Cocaine was the cause of $\tilde{4}$0% of deaths with a support value of 0.398. cocaine and fentanyl had a support metric of 0.291, which means that nearly 1 in 3 deaths during this time period and state was caused by a lethal combination of cocaine and fentanyl. The confidence metric tells us the probability of a specific second drug being present if a different one is already present. The most lethal combination of drugs from this data was Xylazine and fentanyl. With a confidence value of 0.994, there is a 99% chance that if a victim has Xylazine in their system, there will also be fentanyl. Prior to this, we had never heard of Xylazine but after research we discovered that Xylaxine is an animal tranquilizer that is utilized as a cutting agent. Cutting agents are substances that are added to drugs to increase the product's volume, aka, fillers. This combination had a lift of 1.46, meaning that this combination is happening more frequently than it would be by random chance. The second most prevalent drug combination was fentanyl with cocaine. Fentanyl and cocaine had a confidence value of approximately 0.73, meaning that if fentanyl is present, there is a 73% chance that heroin morphine codeine is also. With a lift value of approximately 1.07, this combination is happening much more frequently than it would if it only occurred from random chance.

3.2.2 Regression Selection. For another model, I decided to perform regression to determine whether there are months that have predictive power for drug related deaths and if year has an effect on the number of deaths. This can help provide insight into whether drug related deaths have any seasonality pattern. For my response variable, I had to create a new data frame called 'Deaths' that groups the deaths by the months and years that they occurred and tallies the counts for each month, this will be our response variable.

Assumptions. My first model was just using 'Deaths' as the response with the features, 'Month' and 'Year'. My summary output showed a $R\hat{2}$ value of 0.64, indicating that 64% of the variability in the data is explained by this model. In regression, we need to verify that the linearity assumption and constant variance assumption is intact. For this a fitted versus residual plot was made. After analyzing the plot, both assumptions were violated. The plot showed a parabolic pattern and heteroskedasticity, indicating that the assumptions of linearity have been violated and there was inconsistent variance. Therefore, we decided on another model to face these challenges.

Hyperparameter Tuning. Squaring the 'Year' feature could help address this issue by allowing the model to capture the non-linear relationship between 'Year' and 'Deaths'. Additionally, the variance of the residuals appears to increase as the predicted values increase due to heteroscedasticity. This could be addressed by applying a logarithmic transformation to the dependent variable (Deaths) to stabilize the variance. After adding the squared 'Year' feature and taking the log of 'Deaths', we conducted another residual versus fitted plot. Our new plot shows a random scattering of the data points (homoscedasticity), and there is no pattern, therefore, our linearity assumption and constant variance assumptions are now intact.

Solutions. The new r-squared value is 0.883, meaning that 88.3% of the variability is explained by the model. Compared to our original model, this model with our transformations is a much better fit for the data, whose r-squared value was 0.64.

3.3 Treatment Episode Data Set: Admissions

3.3.1 Classification Selection and Assumptions. A classification model was chosen for the full cleaned version of The Treatment Episode Dataset: Admissions in the form of a decision tree and logistic regression. Specifically the decision tree was chosen as this dataset is highly categorical, and it is designed to handle categorical data splits with ease. Additionally, this does not assume a linear relationship between the features and target and is non-parametric. Decision trees assume that features have predictive power to create meaningful splits in the data. Most importantly, they are highly interpretable; therefore, we analyzed the tree's features to directly see feature importance. The decision tree allows us to look into the strongest underlying risk factors that predict a patient will have a co-occurring mental health diagnosis. The logistic regression was chosen because it is the industry-standard baseline for binary classification. Logistic regression assumes a linear relationship of the logits of mental health diagnosis and the independent variables. After transforming text categories into numerical features, the model implementation was very simple. This allows us to look beyond just one variable at a time; instead, we are able to take a wide variety of different risk factors and calculate the overall probability. Specifically, we are able to assess these risk factors and find the probability of a patient requiring a dual-diagnosis treatment. This makes the logistic regression highly interpretable and time effective for complex datasets.

For the Treatment Episode Dataset: Admissions subset, classification was implemented in the form of a random forest. The random forest does not assume anything about the distribution of the data and is non-parametric. However, it does assume that features have predictive power to make meaningful splits in the data. While regression is generally the default for age-based analyses, age in this case was in discrete brackets rather than continuous, making regression not the ideal choice. The random forest classifier does not rely on simply one decision tree but rather 100 different trees to determine the outcome, mitigating overfitting. This gives us a high level of stability and accuracy when dealing with complex data and many categorical features. This approach gave us the tools to identify key predictors, specifically highlighting how primary substance choice and demographic background interact to influence age of first use patterns.

Hyperparameter Tuning. For all classification modeling for both the TEDS-A and its subset, feature selection and cleaning were conducted, filtering for main predictors. Specifically, dropna() was used on the targets of 'mental_health' and 'first_use' and their primary predictors. This was done to ensure that the models were trained only on complete observations. Additionally, because machine learning algorithms cannot process text, one-hot encoding was implemented using pd.get_dummies to expand features into binary columns. This allowed for the models to find the importance of each category independently. Lastly, StandardScaler was used to transform features into z-scores. This was critically important for the logistic regression to ensure that the variables with larger numerical ranges did not have disproportionate influence. A specific challenge faced for this subset was that initially, linear regression was considered. However, we quickly realized this would not work as the target variables are discrete rather than continuous. To address this challenge, classification was re-framed in a random forest model to capture the complex nuance of this data.

The training process for the complete TEDS-A dataset and its subset was conducted to ensure the models could predict unknown outcomes. The first step was to create a train-test split, where the data was partitioned into 70% for the training set and 30% for the testing set. Additionally, stratification was implemented to ensure the sets had the same proportion of "Yes" vs. "No" for diagnoses. During the model fitting stage, the random forest model utilized 100 independent decision trees by implementing n_estimators = 100 to ensure the "split" points were accurate for 'first_use' prediction. Additionally, the use of max_depth ensured the model identified underlying patterns and did not get lost in the noise. The logistic regression calculated the weights of each variable, helping to determine which features increased or decreased the probability of a diagnosis. A critical part of the training process and tuning was using class_weight = 'balanced', ensuring the model will not just pick the most common age group every time. After the training phase, the testing phase was done by using the testing set in the model to generate predictions and then comparing those with the truth.

Evaluation and Insights. To measure model performance for co-occurring mental diagnoses, we assessed accuracy, precision, recall and ROC AUC curves. Both the decision tree and logistic regression achieved accuracy levels of 60%. Because this test set was highly balanced, the results are more meaningful than a 50/50 random guess. To evaluate the sensitivity compared to specificity, ROC curves were generated. The logistic regression had an AUC of 0.64, which is slightly higher than the AUC of 0.63 for the decision tree. The smooth curve of the logistic regression suggests that the relationship between demographics, substance use, and mental health is relatively linear. This makes the baseline logistic model more reliable for co-occurring mental diagnoses. Due to the use of class_weight = 'balanced', the logistic regression was able to have an almost near perfect balance for recall (0.60 for class 0 and 0.61 for class 1), further proving it is not biased toward either of these outcomes. By examining feature importance using the decision tree, we were able to make critical insights in regard to co-occurring mental diagnoses. The most common single predictor for co-occurring mental diagnoses is 'No Prior Treatment', this explained over 46% of model's decision making weight. In addition, there were many key insights in regards to demographic markers. The labels of being male and having a race of others had 15% and 16% splits in the tree respectively, being the second and third most important. The top substance related for predicting health issues was methamphetamine/Speed at 6%.

The Random Forest Model used to predict 'Age of First Use' was evaluated via multi-class classification report and a chronological confusion matrix. The model had an accuracy of 20.2%. The accuracy across the 7 categories was 14.2%. This indicates that the model is not blindly guessing and rather calculating meaningful statistics. The biggest strength of the model is being able to look into the extreme ages in the spectrum. Because of the prioritization of balancing class weight, the Recall of 0.60 for $\leq$ 11 age group and 0.54 for those 30+. The Chronological confusion matrix also supports the previous results. There is a heavy concentration of accurate predictions for those $\leq$11 and 30+. The top five most important features were all specific substances. The two biggest predictors were Other Opiates (26.3%) and Synthetics and Heroin (20.9%).

3.4 Training Process

4 Results and Discussion

4.1 Evaluation

4.2 Insights

Appendices

A Data Exploration

A.1 NCHS Mortality

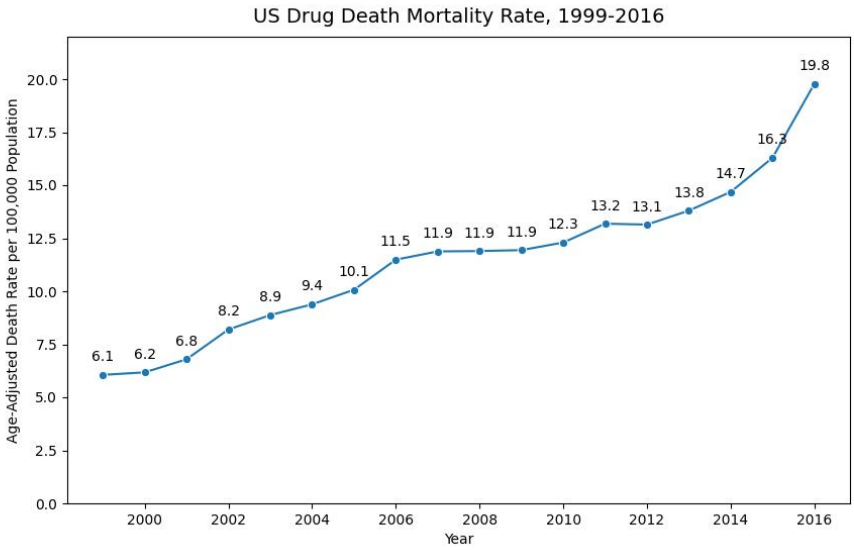

Figure 2: Rise in U.S. drug mortality rates from 1999–2016, including a sharp increase between 2014 and 2016.

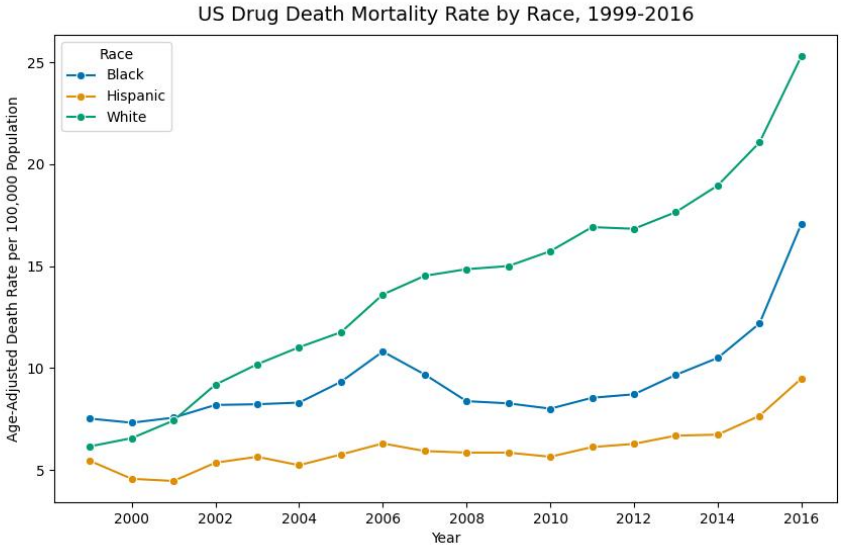

Figure 3: Increase in drug mortality rates from 1999–2016 by race, showing a disproportionate impact on non-Hispanic White populations.

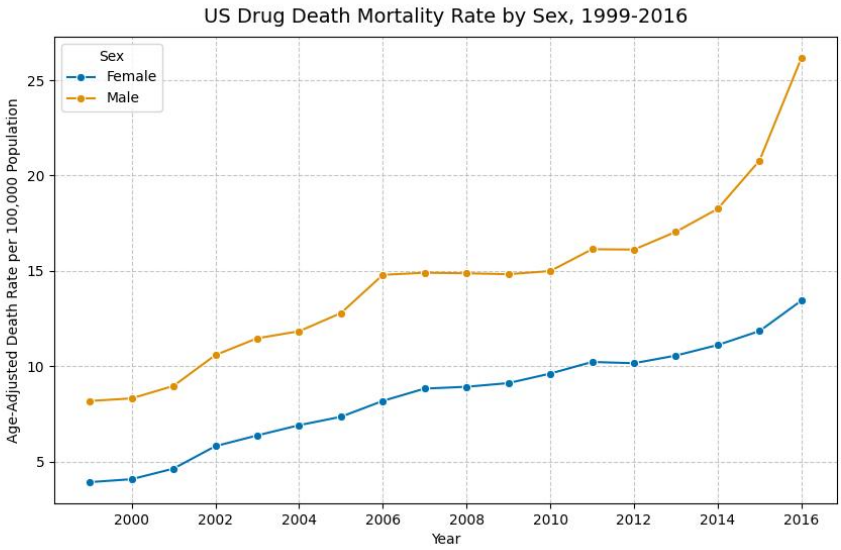

Figure 4: Drug mortality rates by sex from 1999–2016, with consistently higher rates among men than women.

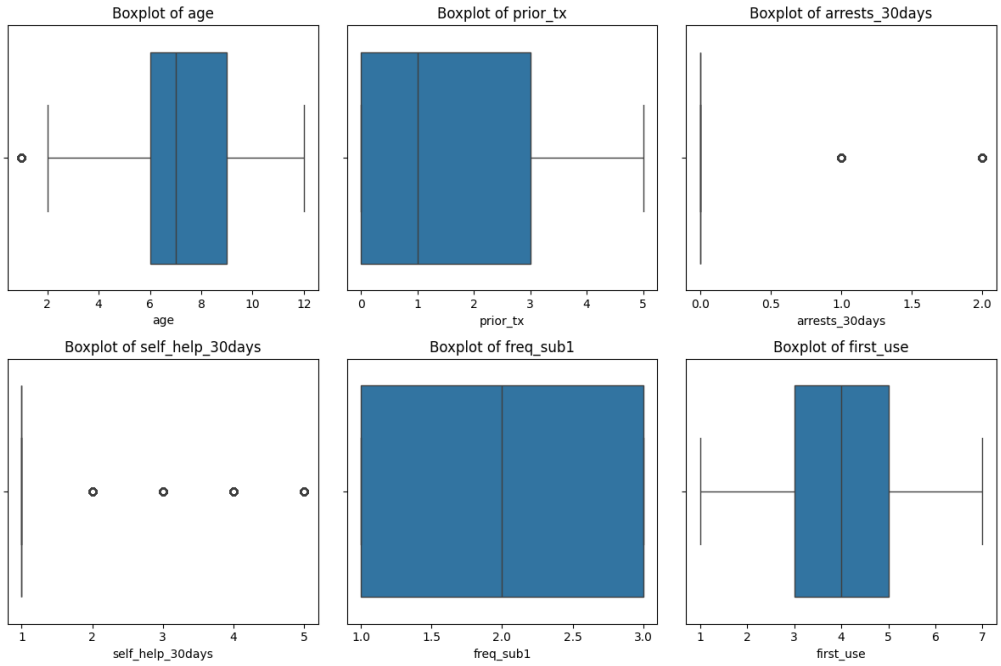

Figure 5: Distribution of drug mortality rates by age group, with the highest rates among individuals aged 25–44.

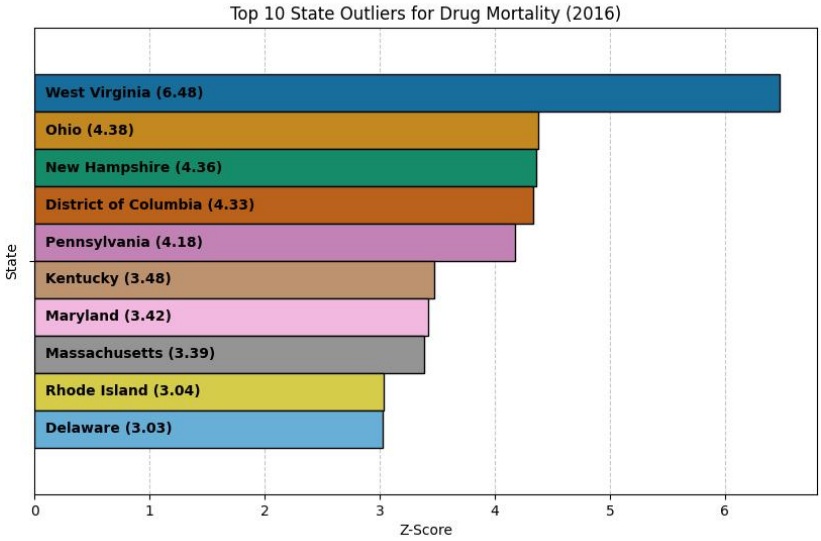

Figure 6: States with the ten highest drug mortality rate outliers in 2016.

A.2 Connecticut

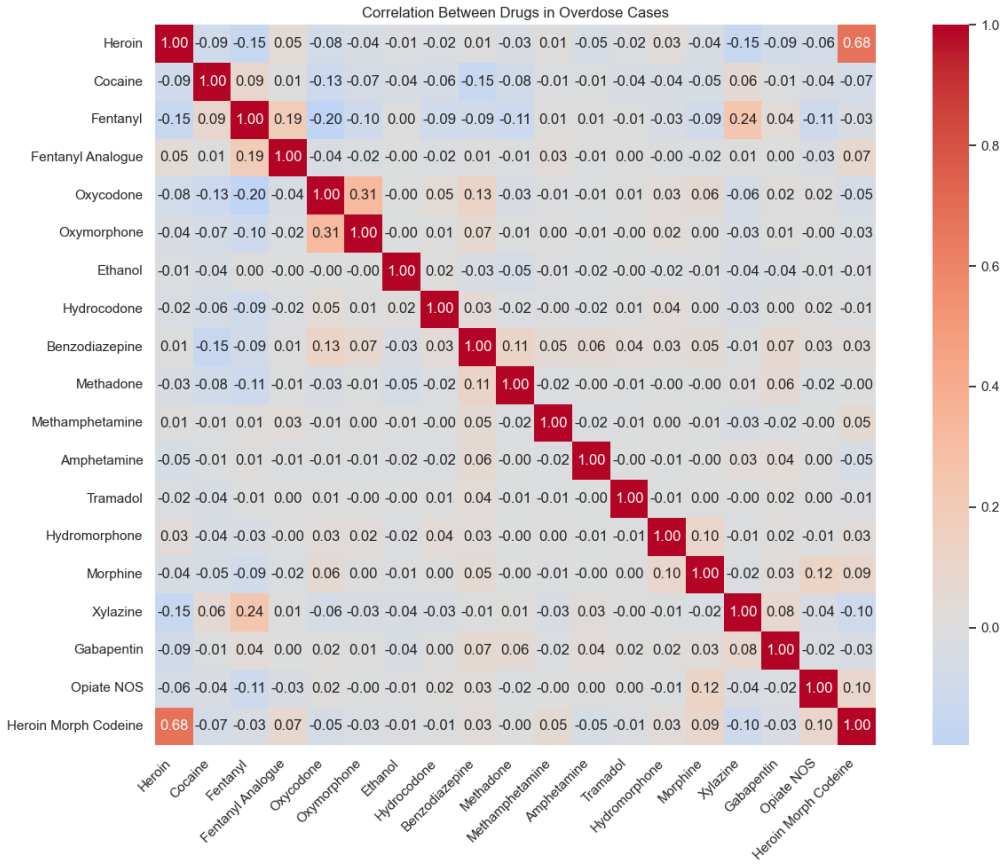

Figure 7: The heatmap shows that there is a weak positive correlation between Xylazine and Fentanyl. There is a strong correlation between Heroin and Heroin Morphine Codeine, meaning that the combination of Heroin Morphine and Codeine combination is prevalent

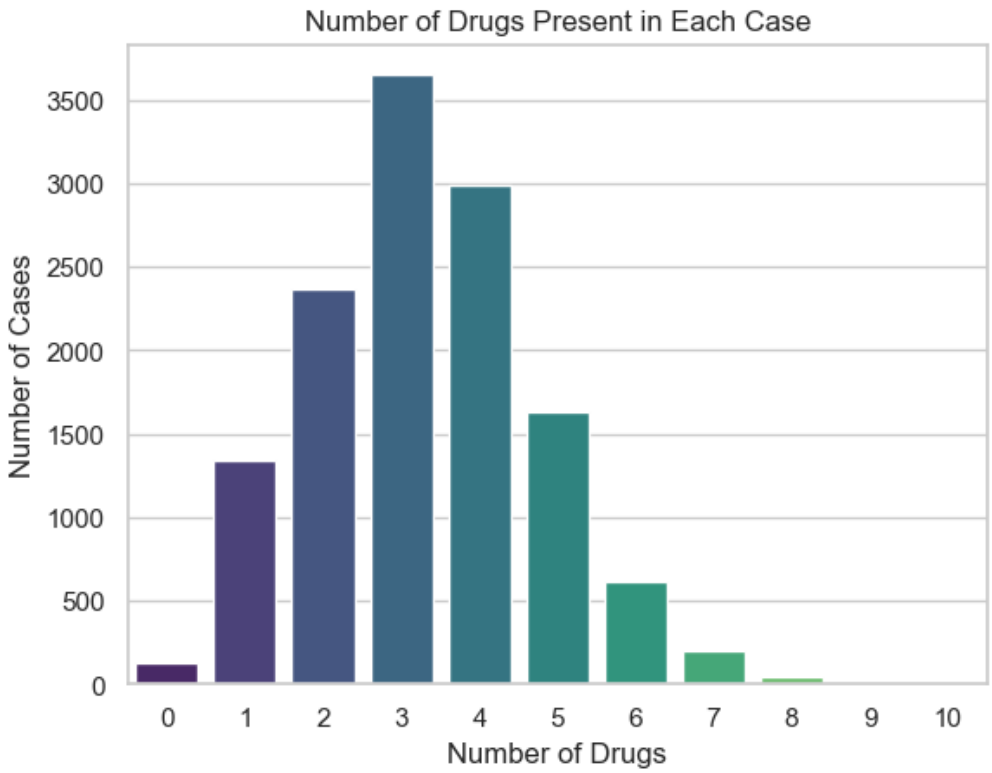

Figure 8: This visual is to compare the transformation of Drug Count prior to the log transformation. From the shape, we can see that the distribution is right-skewed

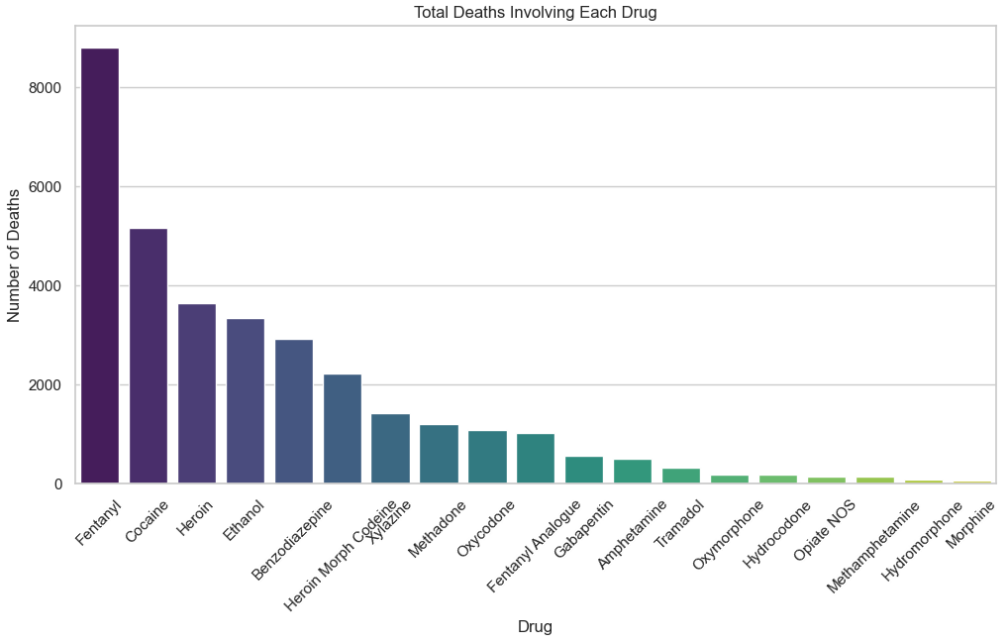

Figure 9: From this bar chart, we can see that the leading drug causing overdoses in Connecticut from 2012 to 2024 is Fentanyl, with Cocaine and Heroin being second and third leading drug

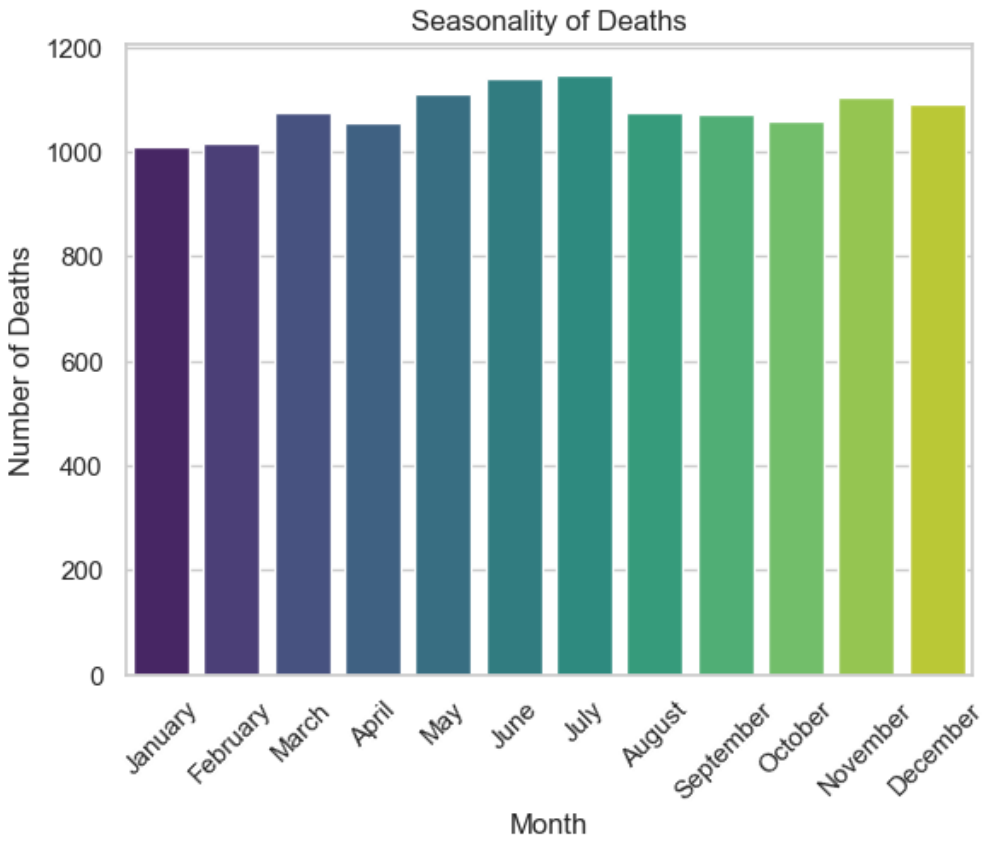

Figure 10: From this bar chart, we can see that the peak summer months, June and July, tend to have slightly higher overdose cases than the other months

A.3 Treatment Episodes

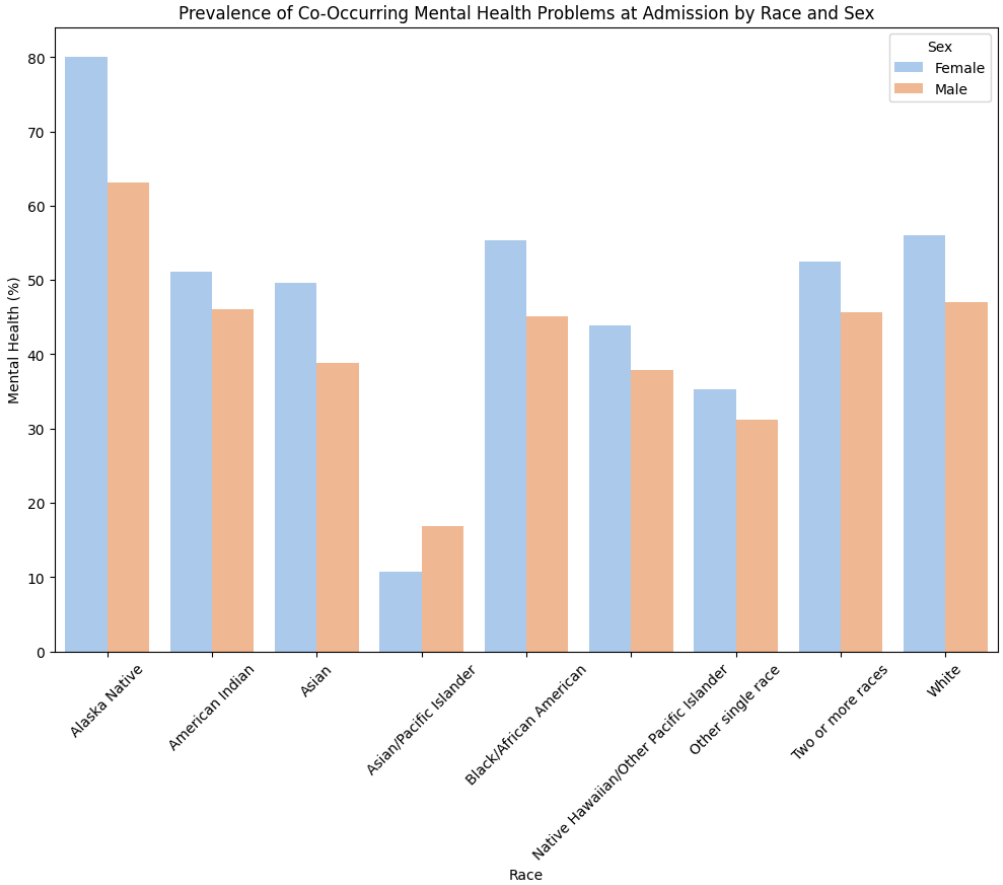

Figure 11: Alaskan Native females seem to have the highest co-occurrence of mental-health problems at admission

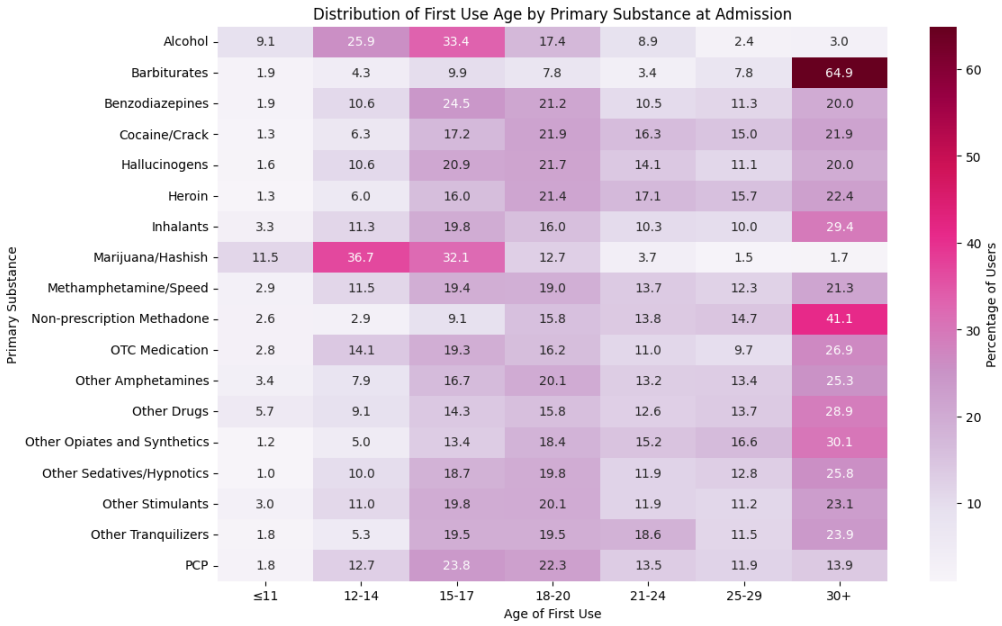

Figure 12: Heatmap of age of first use with primary substance. Those whose first use was 30+ seem to gravitate to barbiturates.

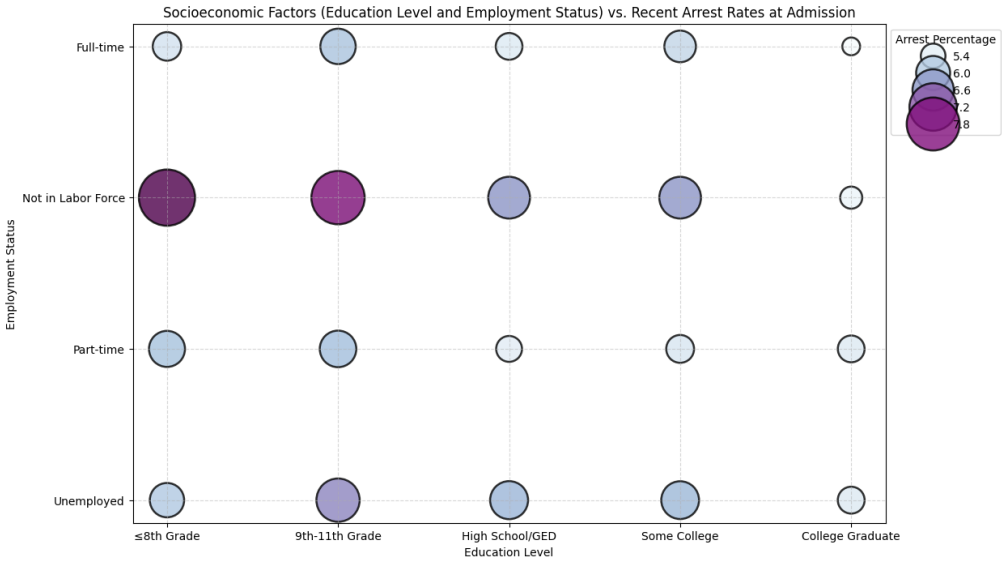

Figure 13: Those with less education while being unemployed seem to have been arrested more within the past 30 days of admission

B Models

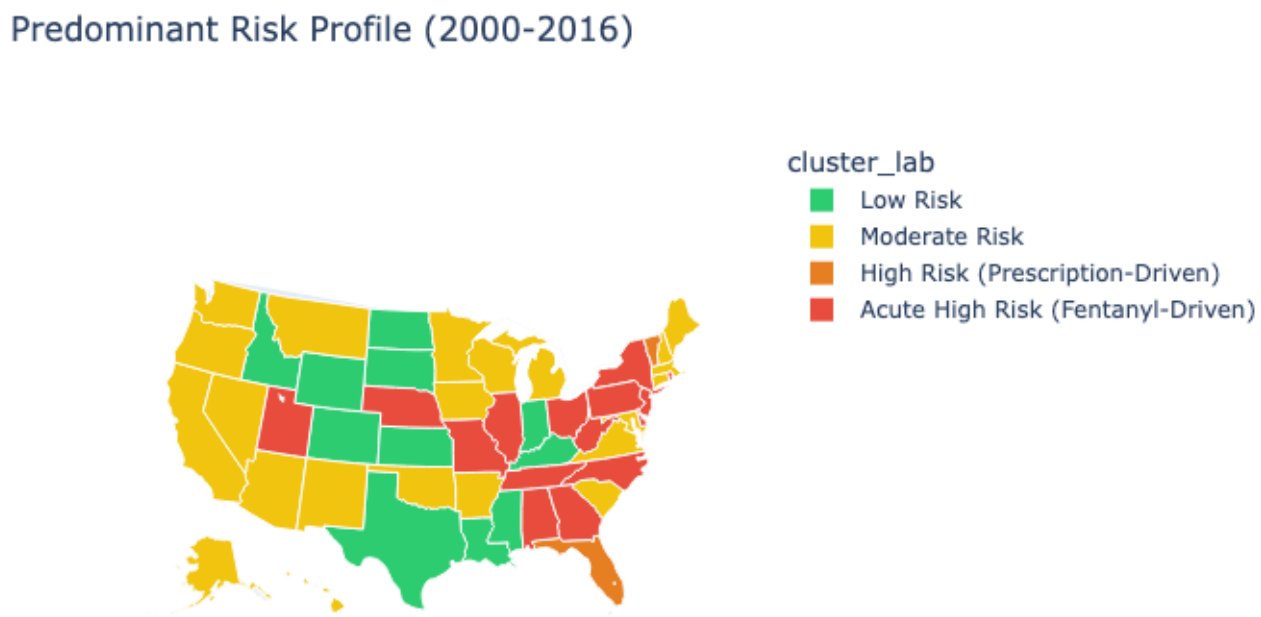

Figure 14: Dominant risk profiles for U.S. states from 2000-2016

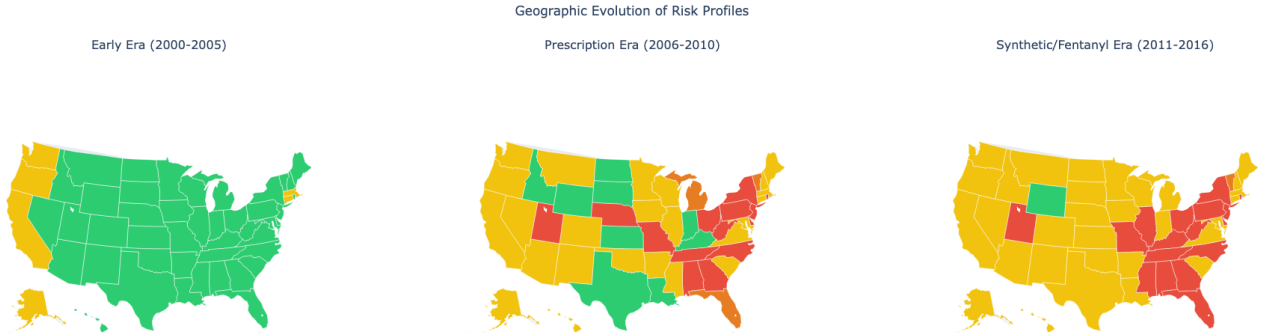

Figure 15: Evolution of State Risk Profiles

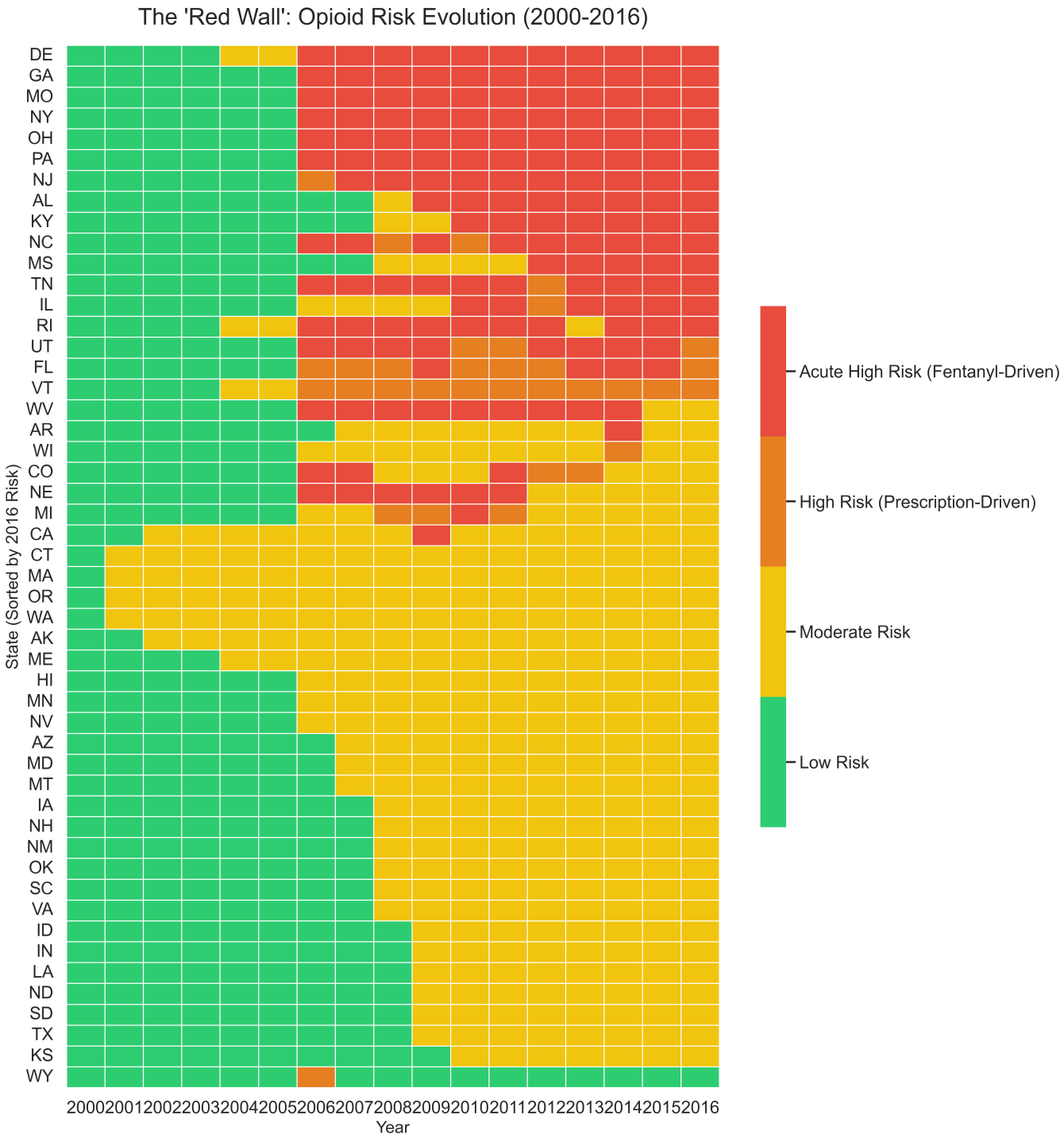

Figure 16: Heatmap: Evolution of State Risk Profiles